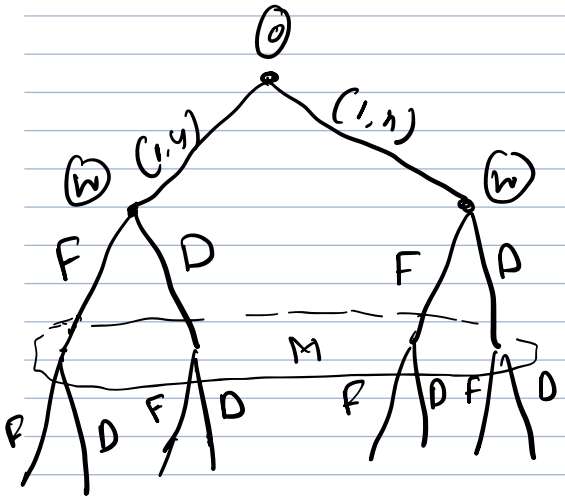
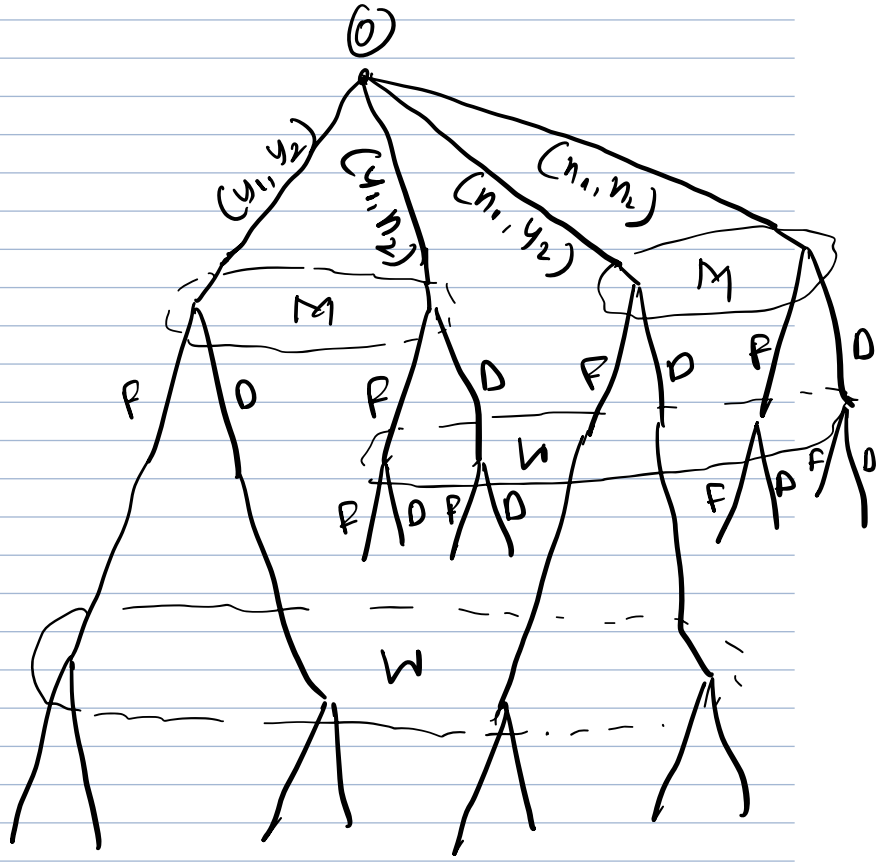
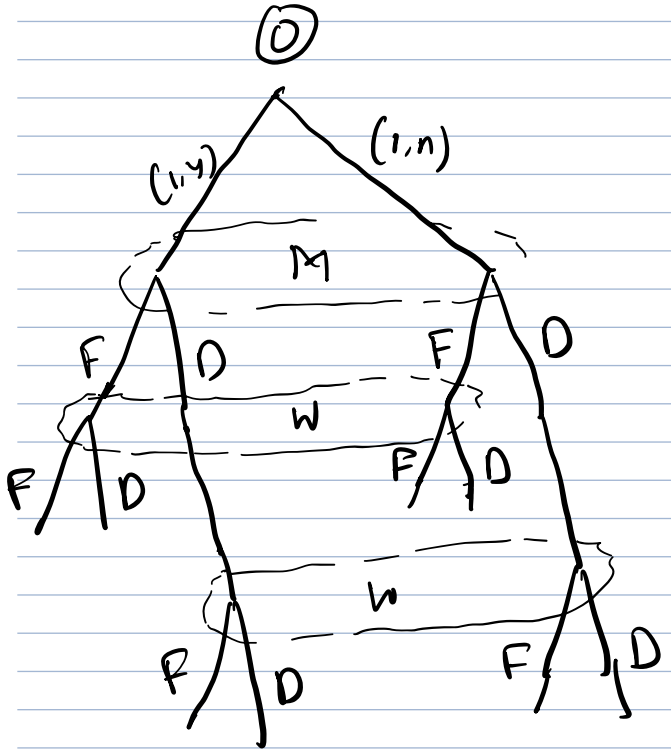
Class 32







Theorem Every Harsanyi Game of incomplete information can be written as an extensive form game with imperfect information and chance nodes.

- For the extensive game form representations, the number of information sets for player i

is $|T_i|$.

- The game will have perfect recall for all the players. This mean we may use behavior strategies.

pure

- A strategy for player i, $s_i : T_i \to \bigcup_{t_i \in T_i} A_i(t_i)$

  where $s_i(t_i) \in A_i(t_i) \quad \forall t_i \in T_i$

- A mixed strategy is a prob distribution over the pure strategies.

- A behavior strategy for player i is $b_i : T_i \to \bigcup_{t_i \in T_i} \Delta A_i(t_i)$

  where $b_i(t_i) \in \Delta A_i(t_i) \quad \forall t_i \in T_i$.

$s_i : H_i \to \bigcup_{h \in H_i} A(h)$

$s_i(h_i) \in A(h_i)$

Consider the strategy profile $b = (b_1, \dots, b_n)$.

For the type profile $t = (t_1, \dots, t_n)$ and an action profile $a = (a_1, \dots, a_n)$ where $a_i \in A_i(t_i) \quad \forall i = 1(1)n$.

For player i, the utility at type profile t is

$$U_i(t; b) = \sum_{a \in \times_{i \in N} A_i(t_i)} b_1(t_1; a_1) \times b_2(a_2; t_2) \dots \times b_n(t_n, a_n)\, u_i(t; a)$$

Finally, the total utility will be

$$U_i(b) = \sum p(t)\, U_i(t; b)$$

$a_1 \in A_1(t_1)$

$a_2 \in A_2(t_2)$

$\vdots$

$\sum_{A \in T}$

$a_i \in A_i(t_i)$

This is called the interim stage, before the randomization takes place.

After player i knows his type t_i, also called interim utility.

$$U_i(b \mid t_i) = \sum_{A_{-i} \in T_{-i}} p(t_{-i} \mid t_i)\, U_i((t_i, A_{-i}), b)$$

$$U_i(b)$$

$$= \sum_{t \in T} p(t)\, U_i(t_i\, b)$$

$$= \sum_{t \in T} p(t_{-i} \mid t_i)\, p(t_i)\, U_i(t_i, b)$$

$$= \sum_{t_i \in T_i} \sum_{t_{-i} \in T_{-i}} p(t_i) \left(p(t_{-i} \mid t_i)\, U_i(t_i, b) \right)$$

$$= \sum_{t_i \in T_i} p(t_i) \sum_{t_{-i} \in T_{-i}} p(t_{-i} \mid t_i)\, U_i(t_i, b)$$

$$= \sum_{t_i \in T_i} p(t_i)\, U(b \mid t_i)$$

- **Nash Equilibrium** A Nash pay profile

$\sigma = (\sigma_i)_{i \in N}$ is an NE if

$$U_i(\sigma^*) \geq U_i(\sigma_i, \sigma_{-i}^*) \quad \forall i \in N$$

and all $\sigma_i \in B_i$

- **Bayesian Nash Equilibrium** A strategy profile σ^* is a BNE if

$$U_i(\sigma^* | t_i) \geq U_i((a_i, \sigma_{-i}^*) | t_i)$$

$\forall i \in N$

$\forall t_i \in T_i$

$\forall a_i \in A_i(t_i)$.

t_1, t_2, t_3

$$U_i(\sigma) = p(t_1) U_i(\sigma | t_1) + p(t_2) U_i(\sigma | t_2) + p(t_3) U_i(\sigma | t_3)$$

Result. Every BNE is also an NE.

Proof: See Solan's book.

Theorem For an incomplete information game, an NE exists in behavior strategies.

Proof: The corresponding Extensive form game has a NE in mixed strategies. And, a

the game has perfect recall, every mixed strategy is equivalent to a behavior strategy, which means a behavior strategy NE exists.